

Make Figures

Load libraries

```
# Load the required libraries
library(ggplot2)
library(cowplot)
library(egg)
```

```
## Loading required package: gridExtra
```

```
library(xtable)

# Load the required functions
z = source("codes/helpers.R")
image_path = "images/"
results_path = "results/"
```

Explanatory figure

```
p1 <- ggdraw() + draw_image(paste(image_path, "sdproc.svg", sep=""), scale = 0.9)
```

```
## Warning: Package 'magick' is required to draw images. Image not drawn.
```

```
p2 <- ggdraw() + draw_image(paste(image_path, "arrow.svg", sep=""), scale = 0.9)
```

```
## Warning: Package 'magick' is required to draw images. Image not drawn.
```

```
# Use `plot_grid` to assemble figure 1 from the base figures.
f1 <- plot_grid(p1, p2, labels = c("a", "b"))
save_plot(paste(image_path, "fig1.png", sep=""), f1, ncol=2)
# This file is not used anymore to make fig1. Directly edit the SVG instead.
```

Fitting figure

Preliminaries

Load the data and define some plotting parameters.

```
load(paste(results_path, "fit.Rda", sep=""))
# Plot parameters
lwidth = 2; ht = 0.86; bwidth = 2.5; wd2=0.47
lty1 = 1; lty2 = 6;
cex_plot = 1.4; cex_text = 1.5; cex_data = 1.7;
c1 = "#1b9e77bb";
```

Plot and assemble figure

```
set.seed(0) # Set seed so we the jitter is reproducible
# Comment out accordingly for jpeg or pdf
# jpeg(paste(image_path, "fit_both.jpeg", sep=""),
#       width = 8, height = 4.2, units = 'in', res = 300)
pdf(paste(image_path, "fit_both.pdf", sep=""), width = 8, height = 4.2)

# Create the layout of the plot. There are 3 rows and 2 columns.
# The top row consists of the y label, dataset 1 fits and dataset fits.
# The bottom row is entirely allocated for the legend.
m <- matrix(c(1,2,3,4,4,4), 2, 3, byrow=TRUE)
layout(mat=m, heights=c(ht,1-ht), widths=c(1-2*wd2, wd2, wd2))

# Create the y axis label
par(mar=c(0,0,0,0))
plot(1, type = "n", axes=F, xlab="", ylab="")
text(0.8,1.05,"P(d)",srt=90, cex=cex_plot)

# Retrieve exponential model fit
dataset_no = 1;
ds = getdat(dataset_no)
r = results$r[dataset_no]
alpha = results$alpha[dataset_no]
beta = results$beta[dataset_no]
dose = 10^seq(log10(min(ds$dose)), log10(max(ds$dose)), 0.02) # Define the dose

# Plot the first dataset
par(mar=c(4.0,0.9,0.5,0.5)) # Set the margin for this plot and the next
## Plot the data
plot(jitter(log10(ds$dose), factor=0.1), ds$null/ds$ntot,
      xlab=expression(Log[10]~(dose)),
      ylim=c(-0,1), bg = c1, pch = 21, cex=cex_data,
      cex.axis = cex_plot, cex.lab=cex_plot);
## Plot the two fits
lines(log10(dose), 1-exp(-r*dose), lwd=lwidth, lty=lty1)
lines(log10(dose), 1-(1+dose/beta)^(-alpha), lwd=lwidth, lty=lty2)
box(lwd=bwidth)
text(4.0, 0.95, expression(bold("a")), cex=cex_text)

# Retrieve beta Poisson model fit
dataset_no = 2;
ds = getdat(dataset_no)
r = results$r[dataset_no]
```

```

alpha = results$alpha[dataset_no]
beta = results$beta[dataset_no]
dose = 10^seq(log10(min(ds$dose)), log10(max(ds$dose)), 0.02) # Define the dose

# Plot the second dataset
## Plot the data
plot(jitter(log10(ds$dose), factor=0.05), ds$null/ds$ntot,
     xlab=expression(Log[10]~(dose)), ylab = "",
     ylim=c(-0,1), bg = c1, pch = 21, cex=cex_data,
     cex.axis = cex_plot, cex.sub=cex_plot, cex.lab=cex_plot, yaxt='n')
## Plot the two fits
lines(log10(dose), 1-exp(-r*dose), lwd=lwidth, lty=lty1)
lines(log10(dose), 1-(1+dose/beta)^(-alpha), lwd=lwidth, lty=lty2)
box(lwd=bwidth)
text(6.0, 0.95, expression(bold("b")), cex=cex_text)

# Plot the legend
par(mar=c(0,0.1,0,1), lwd=bwidth) # Reset the margin
plot(1, type = "n", axes=F, xlab="", ylab="") # Create empty plot
legend("topleft", c("Data", "Exponential", "Beta-Poisson"), pch=c(21, NA, NA),
      pt.bg=c(c1, NA, NA), pt.lwd=c(1,NA,NA), pt.cex = cex_data,
      lty=c(NA, lty1, lty2),lwd=c(NA, lwidth, lwidth),
      inset = 0.25, cex=cex_plot, horiz = TRUE, bg="transparent")

dev.off()

```

```

## pdf
## 2

```

View the dose response data

Print out the fitting results in a LaTeX table.

```

dataset_no = 1
ds = getdat(dataset_no)
df_data = data.frame(Dose = ds$dose, Null = ds$null, Ntot = ds$ntot)
xtable(df_data, display=c("d", "e", "d", "d"))

```

```

## % latex table generated in R 4.5.3 by xtable 1.8-8 package
## % Thu May 28 16:17:12 2026
## \begin{table}[ht]
## \centering
## \begin{tabular}{rrrr}
## \hline
## & Dose & Null & Ntot \\
## \hline
## 1 & 1.00e+04 & 0 & 5 \\
## 2 & 1.00e+04 & 0 & 5 \\
## 3 & 1.00e+06 & 0 & 5 \\
## 4 & 1.00e+06 & 1 & 9 \\
## 5 & 1.00e+08 & 5 & 8

```

```
## 6 & 1.00e+08 & 3 & 5 \\
## \hline
## \end{tabular}
## \end{table}
```

```
dataset_no = 2
ds = getdat(dataset_no)
df_data = data.frame(Dose = ds$dose, Nill = ds$null, Ntot = ds$ntot)
xtable(df_data, display=c("d", "e", "d", "d"))
```

```
## % latex table generated in R 4.5.3 by xtable 1.8-8 package
## % Thu May 28 16:17:12 2026
## \begin{table}[ht]
## \centering
## \begin{tabular}{rrrr}
## \hline
## & Dose & Nill & Ntot \\
## \hline
## 1 & 1.00e+06 & 0 & 4 \\
## 2 & 1.00e+06 & 1 & 5 \\
## 3 & 1.00e+08 & 1 & 5 \\
## 4 & 5.00e+08 & 3 & 5 \\
## 5 & 2.50e+09 & 6 & 6 \\
## 6 & 1.00e+10 & 9 & 10 \\
## 7 & 1.00e+10 & 9 & 14 \\
## 8 & 1.00e+10 & 3 & 5 \\
## 9 & 1.00e+10 & 5 & 5 \\
## 10 & 2.00e+10 & 2 & 2 \\
## 11 & 2.30e+10 & 14 & 19 \\
## \hline
## \end{tabular}
## \end{table}
```

Effect of fr and C

Preliminaries

```
load(paste(results_path, "effect.Rda", sep=""))
sz = 11; sz_ax = 14; sz_leg = 14; sz_lab = 6;
```

Plot and assemble figure

Make the plots and store them in handles. Then use `grid_extra` from the `cowplot` package to assemble them.

```
# Exponential, concentration
llim = -7.8
df = top[[1]]
plot1 <- ggplot(df, aes(dose)) +
```

```

geom_line(aes(y=risk, color=C), size=1.5)+
geom_point(aes(y=risk, color=C, shape=abfail), size=4, alpha=0.9)+
xlab("")+
ylab(expression(Log[10]~(P[illness])))+
labs(color=expression('%~MIC), shape="Antibiotic")+theme_bw(base_size = sz)+
theme(
  legend.position="top",
  legend.box="vertical",
  axis.line = element_line(size=1),
  axis.text = element_text(face="bold", size=sz_ax),
  axis.title = element_text(size=sz_ax),
  axis.text.x = element_blank(),
  legend.text = element_text(size=sz_leg),
  legend.title = element_text(size=sz_leg),
  plot.background = element_blank())+
guides(shape = guide_legend(direction = "horizontal", order=2),
  color = guide_legend(direction = "horizontal", order=1))+
scale_y_continuous(limits = c(llim, 0.5))+
annotate("text", x=2, y=0, label="a", size=sz_lab, fontface="bold")

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

```

## Warning: The 'size' argument of 'element_line()' is deprecated as of ggplot2 3.4.0.
## i Please use the 'linewidth' argument instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

```

# Exponential, fr
df2 = top[[2]]
plot2 <- ggplot(df2, aes(dose)) +
  geom_line(aes(y=risk, color=fr), size=1.5)+
  geom_point(aes(y=risk, color=fr, shape=abfail), size=4, alpha=0.9)+
  # xlab(expression(Log[10]~(dose)))+
  xlab("")+
  ylab("")+
  labs(color=expression(f[r]), shape=expression(Antibiotic))+theme_bw(base_size = sz)+
  theme(legend.position="top",
    legend.box="vertical",
    axis.line = element_line(size=1),
    axis.text = element_text(face="bold", size=sz_ax),
    axis.title = element_text(size=sz_ax),
    axis.text.x = element_blank(),
    axis.text.y = element_blank(),
    legend.text = element_text(size=sz_leg),
    legend.title = element_text(size=sz_leg),
    plot.background = element_blank())+
  guides(shape = guide_legend(direction = "horizontal", order=2),
    color = guide_legend(direction = "horizontal", order=1))+

```

```

scale_y_continuous(limits = c(llim, 0.5))+
annotate("text", x=2, y=0, label="b", size=sz_lab, fontface="bold")

# Beta Poisson, concentration
df = bottom[[1]]
plot3 <- ggplot(df, aes(dose)) +
  geom_line(aes(y=riskbp, color=C), size=1.5)+
  geom_point(aes(y=riskbp, color=C, shape=abfailbp), size=4, alpha=0.9)+
  xlab(expression(Log[10]~(dose)))+
  ylab(expression(Log[10]~(P[illness])))+
  labs(color=expression(Ab.~Conc.~mg.L^{-1}), shape="Antibiotic")+theme_bw(base_size = sz)+
  # ggtitle(plot_title)+
  theme(legend.position="none",
        axis.line = element_line(size=1),
        axis.text = element_text(face="bold", size=sz_ax),
        axis.title = element_text(size=sz_ax),
        legend.text = element_text(size=sz_leg),
        legend.title = element_text(size=sz_leg))+
  # annotate("text", x=5, y=-12, label="<same legend as above>", size=4.5)+
  scale_y_continuous(limits = c(llim, 0.5))+
  annotate("text", x=2, y=0, label="c", size=sz_lab, fontface="bold")

# Beta Poisson, fr
df2 = bottom[[2]]
plot4 <- ggplot(df2, aes(dose)) +
  geom_line(aes(y=riskbp, color=fr), size=1.5)+
  geom_point(aes(y=riskbp, color=fr, shape=abfailbp), size=4, alpha=0.9)+
  xlab(expression(Log[10]~(dose)))+
  ylab("")+
  labs(color=expression(f[r]), shape="Antibiotic")+theme_bw(base_size = sz)+
  theme(legend.position="none",
        axis.line = element_line(size=1),
        axis.text = element_text(face="bold", size=sz_ax),
        axis.title = element_text(size=sz_ax),
        axis.text.y = element_blank(),
        legend.text = element_text(size=sz_leg),
        legend.title = element_text(size=sz_leg),
        plot.background = element_blank())+
  # annotate("text", x=5, y=-12, label="<same legend as above>", size=4.5)+
  scale_y_continuous(limits = c(llim, 0.5))+
  annotate("text", x=2, y=0, label="d", size=sz_lab, fontface="bold")

# Assemble the plots together and save
ggsave(paste(image_path, "effect.pdf", sep=""),
       arrangeGrob(grobs=lapply(list(plot1,
                                     plot2,
                                     plot3,
                                     plot4),
                               set_panel_size,
                               width=unit(10, "cm"),
                               height=unit(9.5, "cm"))),
       width=23.7, height=25, units="cm")

```

Sensitivity figure

Preliminaries

```
load(paste(results_path, "sens.Rda", sep=""))
# Define the x axis labels for each sub plot
parnames1 = c('C', expression(f[r]), 'd', expression(E[max]), 'EC50', 'r',
              expression(t[fs]))
parnames2 = c('C', expression(f[r]), 'd', expression(E[max]), 'EC50',
              expression(alpha), expression(beta), expression(t[fs]))
parnames3 = c('C', expression(f[r]), expression(E[max]), 'EC50', 'r',
              expression(t[fs]))
parnames4 = c('C', expression(f[r]), expression(E[max]), 'EC50',
              expression(alpha), expression(beta), expression(t[fs]))
```

Plot and assemble figure

```
# Plot parameters
lwidth = 5; ht = 0.5; wd=0.5;bwidth = 1.5;
c1 = "#1b9e7777"; c2 = "#d95f0277"; c3 = "#7570b3"; set.seed(0)
cex_plot = 1.3; cex_txt = 1.5; rot = 30;

# jpeg(paste(image_path, "sensitivity.jpeg", sep=""),
#       width = 8, height = 8, units = 'in', res = 300)
pdf(paste(image_path, "sensitivity.pdf", sep=""),
    width = 8, height = 8)

# Create the layout
m <- matrix(c(1,2,3,4), ncol=2, byrow=TRUE)
layout(mat=m, heights = c(ht,1-ht), widths=c(wd, 1-wd))

# Exponential with dose
par(mar = c(4,4,0.5,0), lwd=bwidth)
boxplot(t(res_exp$KS), ylim = c(0, 1), ylab="PAWN index", outline=F,
        cex.axis = cex_plot, cex.sub=cex_plot, cex.lab=cex_plot, xaxt="n")
axis(1, labels=FALSE)
text(x=seq_along(parnames1), y=-0.1, srt = rot, adj= 1, xpd = TRUE, labels = parnames1 , cex=cex_plot)
xvals = rep(seq(length(parnames1)), n); yvals = as.vector(res_exp$KS)
points(jitter(xvals,factor=1.5), yvals, bg=c1, pch=21)
text(0.7, 0.95, expression(bold("a")), cex=cex_txt)
lines(c(0,10), c(critval, critval), lty=2)

# Beta Poisson with dose
par(mar=c(4,4,0.5,0.2), lwd=bwidth)
boxplot(t(res_beta$KS), ylim = c(0, 1), ylab = "", outline=F,
        cex.axis = cex_plot, cex.sub=cex_plot, cex.lab=cex_plot, xaxt="n", xlab="")
axis(1, at=seq_along(parnames2), labels=FALSE)
text(x=seq_along(parnames2), y=-0.1, srt = rot, adj= 1, xpd = TRUE, labels = parnames2 , cex=cex_plot)
xvals = rep(seq(length(parnames2)), n); yvals = as.vector(res_beta$KS)
points(jitter(xvals,factor=1.5), yvals, bg=c1, pch=21)
```

```

text(0.7, 0.95, expression(bold("b")), cex=cex_txt)
lines(c(0,10), c(critval, critval), lty=2)

# Exponential without dose
par(mar=c(4,4,0.5,0), lwd=bwidth)
boxplot(t(resnd_exp$KS), ylim = c(0, 1), ylab="PAWN index", outline=F,
        cex.axis = cex_plot, cex.sub=cex_plot, cex.lab=cex_plot, xaxt="n")
axis(1, labels=FALSE)
text(x=seq_along(parnames3), y=-0.1, srt = rot, adj= 1, xpd = TRUE, labels = parnames3 , cex=cex_plot)
xvals = rep(seq(length(parnames3)), n); yvals = as.vector(resnd_exp$KS)
points(jitter(xvals, factor=1.5), yvals, bg=c2, pch=21)
text(0.7, 0.95, expression(bold("c")), cex=cex_txt)
lines(c(0,10), c(critval, critval), lty=2)

# Beta Poisson without dose
par(mar=c(4,4,0.5,0.2), lwd=bwidth)
boxplot(t(resnd_beta$KS), ylim = c(0, 1), outline=F,
        cex.axis = cex_plot, cex.sub=cex_plot, cex.lab=cex_plot, xaxt="n")
axis(1, labels=FALSE)
text(x=seq_along(parnames4), y=-0.1, srt = rot, adj= 1, xpd = TRUE, labels = parnames4 , cex=cex_plot)
xvals = rep(seq(length(parnames4)), n); yvals = as.vector(resnd_beta$KS)
points(jitter(xvals, factor=1.5), yvals, bg=c2, pch=21)
text(0.7, 0.95, expression(bold("d")), cex=cex_txt)
lines(c(0,10), c(critval, critval), lty=2)

dev.off()

```

```

## pdf
## 2

```

Verify beta procedure

Plot the $\mu_{s,AB}$ CDF and the corresponding predicted beta CDF.

Preliminaries

```

load(paste(results_path, "betaver.Rda", sep=""))

color_list = seq(4)
color_list[2] = "#a6cee3"; color_list[1] = "#1f78b4"; color_list[3] = "#b2df8a";
color_list[4] = "#33a02c"; lwidth = 2;

```

Plot and assemble figure

```

pdf(paste(image_path, "beta_verify.pdf", sep=""),
    width = 8, height = 4)

```



```

# Define the layout
par(mfrow=c(1,2))

# Sub plot A : mu vs density for different concentrations
## Calculate density, define the label and plot the density
d = density(b_res$mu)
legend_list = rbind("0% MIC")
plot(d$x, d$y, lwd = 2, col = color_list[1], type="l",
      xlab=expression(paste(mu)), ylab="Density")
## Loop over other concentrations and repeat
for (ind in seq(2, nrow(b_res$mu_s))) {
  d = density(b_res$mu_s[ind,])
  legend_list = rbind(legend_list, stringr::str_glue(b_res$pMIC[ind]*100, "% MIC"))
  lines(d$x, d$y, lwd = 2, col = color_list[ind])
}
legend("topright", legend_list, lty=c(1,1,1,1),
      lwd=c(lwidth, lwidth, lwidth, lwidth),
      col=c(color_list))
# Label position should be at approximately 0.95
label_position = (par("usr")[3] + par("usr")[4])*0.95
text(4.5, label_position, expression(bold("A"))) # sub plot label

# Sub plot B
x = seq(-25, 0, 0.5)
empcdf = ecdf(b_res$r) # Empirical CDF function
## Analytical prediction at 0 % MIC
plot(x, pbeta(10^x, b_res$alpha, b_res$beta), type="l", lwd=2,
      xlab=expression(Log[10]~(r)), ylab="CDF")
## Empirical CDF from samples
points(x, empcdf(10^x), col = color_list[1])
## Analytical CDF from parameters obtained by fitting samples
lines(x, pbeta(10^x, b_res$alpha_s[1], b_res$beta_s[1]),
      col=color_list[1], lty=3, lwd = lwidth)
legend_list2 = rbind("Analytical", "Empirical", "Fitted")
for (ind in seq(2, nrow(b_res$mu_s))) {
  ## Empirical CDF from samples
  empcdf = ecdf(b_res$r_s[ind,])
  points(x, empcdf(10^(x)), col=color_list[ind])
  ## Analytical CDF from parameters obtained by fitting samples
  lines(x, pbeta(10^x, b_res$alpha_s[ind], b_res$beta_s[ind]),
        col=color_list[ind], lty=2, lwd = lwidth)
}
legend("bottomright", legend_list2, pch=c(NA, 1, NA),
      lty=c(1, NA, 2),
      col=c("#000000", "#000000", "#000000"),
      lwd=c(2, NA, lwidth), inset = 0.0)
text(-23.5, 0.95, expression(bold("B"))) # sub plot label
dev.off()

```

```

## pdf
## 2

```